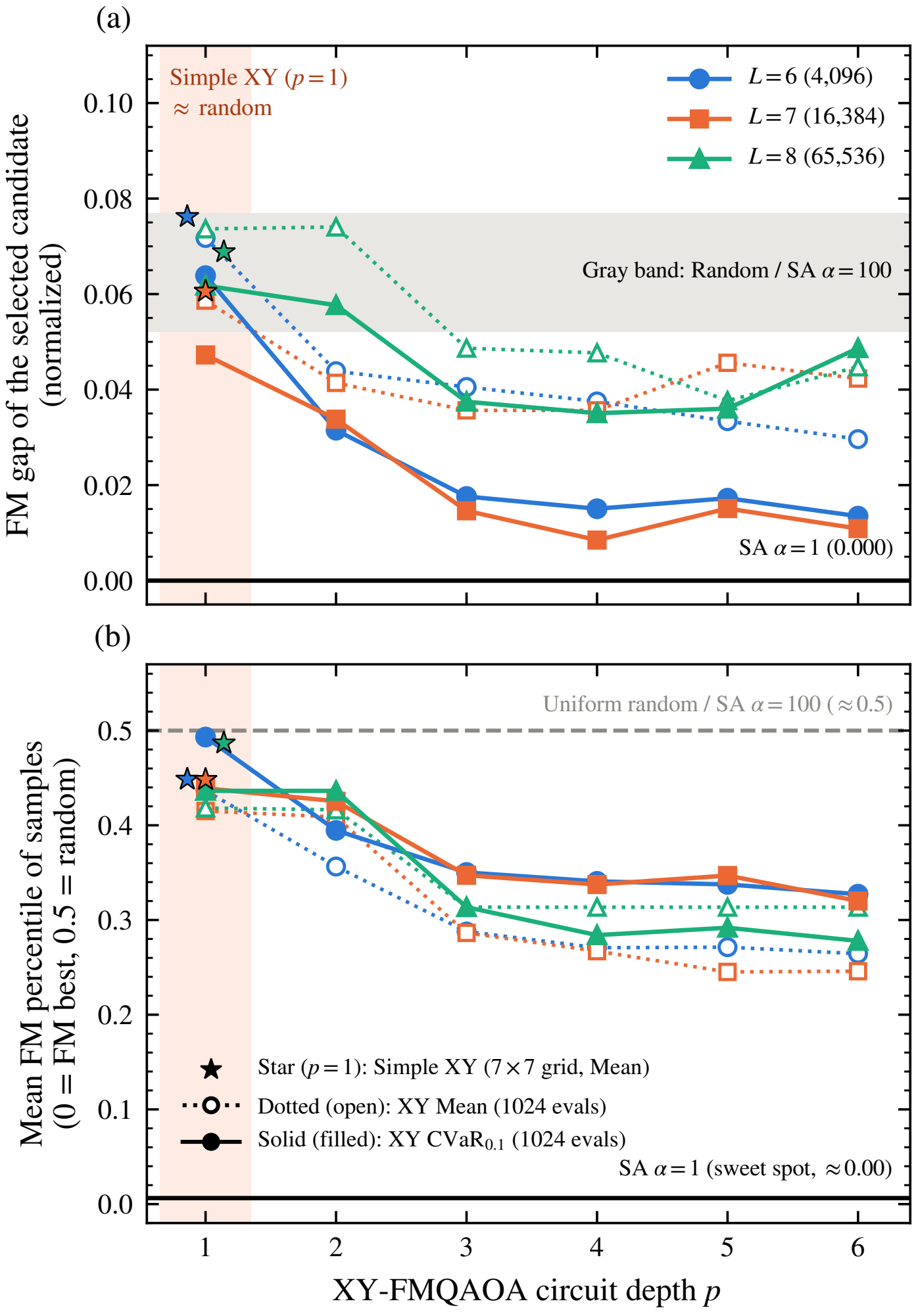
(a)
Simple XY ($p=1$)
≈ random
$L=6$ (4,096)
$L=7$ (16,384)
$L=8$ (65,536)
Gray band: Random / SA $\alpha=100$
SA $\alpha=1$ (0.000)
FM gap of the selected candidate (normalized)
0.10
0.08
0.06
0.04
0.02
0.00
(b)
Uniform random / SA $\alpha=100$ (≈ 0.5)
Star ($p=1$): Simple XY (7×7 grid, Mean)
Dotted (open): XY Mean (1024 evals)
Solid (filled): XY CVaR$_{0.1}$ (1024 evals)
SA $\alpha=1$ (sweet spot, ≈ 0.00)
Mean FM percentile of samples (0 = FM best, 0.5 = random)
0.5
0.4
0.3
0.2
0.1
0.0
1
2
3
4
5
6
XY-FMQAOA circuit depth p